

Tutorial - 5

Q1: Let A be an $n \times n$ matrix over F . For any $\alpha \in F$,
Show that $\det(\alpha A) = \alpha^n \det(A)$.

Proof: Let $A \in M_{n \times n}(F)$, $\alpha \in F$

Let $A = (a_1, a_2, \dots, a_n)$ where $a_i \in F^n \forall i$

Then $\alpha A = (\alpha a_1, \alpha a_2, \dots, \alpha a_n) = (\alpha a_1, \alpha a_2, \dots, \alpha a_n)$

$$\therefore \det(\alpha A) = D(\alpha a_1, \alpha a_2, \dots, \alpha a_n)$$

Now using multilinearity of D for first component, we get.

$$\det(\alpha A) = \alpha D(a_1, \alpha a_2, \dots, \alpha a_n)$$

In this way using multilinearity of D again and again we get.

$$\begin{aligned} \det(\alpha A) &= \alpha \cdot \alpha \cdot \dots \cdot D(a_1, a_2, \dots, a_n) \\ &= \alpha^n \det(A) \end{aligned}$$

□

Q2: An $n \times n$ matrix A over F is skew-symmetric if $A^T = -A$.

If A is a skew symmetric $n \times n$ matrix with complex entries, and n is odd, show that $\det(A) = 0$

Sol:

$$A^T = -A$$

$$\therefore \det(A^T) = \det(A)$$

$$\therefore \det(-A) = \det(A)$$

$$\Rightarrow (-1)^n \det(A) = \det(A)$$

$$(\because \det(kA) = (k)^n \det(A))$$

$$\Rightarrow -\det(A) = \det(A)$$

$$\Rightarrow 2\det(A) = 0$$

$$\Rightarrow \det(A) = 0$$

Q4: An $n \times n$ matrix over F is orthogonal if $AA^T = I$. If A is orthogonal show that $\det(A) = \pm 1$.
Give an example of orthogonal matrix with $\det(A) = -1$.

Sol: $A \in M_{n \times n}(F)$
 $AA^T = I$

$$\Rightarrow \det(AA^T) = \det(I)$$

$$\Rightarrow \det(A)\det(A^T) = 1$$

$$\Rightarrow \det(A)\det(A) = 1 \quad (\because \det(A^T) = \det(A))$$

$$\Rightarrow (\det(A))^2 = 1$$

$$\Rightarrow \det(A) = \pm 1$$

Consider $N = (-e_1, e_2, \dots, e_n) \in M_{n \times n}(F)$

$$\text{i.e. } N = \begin{pmatrix} -1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{pmatrix}$$

Then $N^T = N$

Also $N^2 = I$

$\therefore N N^T = N^T N = I$ so N is orthogonal.

Also, $\det(N) = D(-e_1, e_2, \dots, e_n) = -D(e_1, e_2, \dots, e_n) = -1$.

Q5: Let S_n be the permutation group of n -symbols.

Show that $\text{sgn}: S_n \rightarrow \{\pm 1\}$ defined by $\sigma \mapsto \text{sgn}(\sigma)$ is a group homo. that is $\text{sgn}(\sigma_1 \sigma_2) = \text{sgn}(\sigma_1) \text{sgn}(\sigma_2)$.
 $\forall \sigma_1, \sigma_2 \in S_n$

Sol Define $\text{sgn}: S_n \rightarrow \{ \pm 1 \}$ as $\sigma \rightarrow \text{sgn}(\sigma)$

Let $\sigma_1, \sigma_2 \in S_n$

As every permutation in S_n can be written as product of transposition.

So $\exists$ τ_i 's and τ_j 's s.t.

$$\sigma_1 = \tau_1 \tau_2 \dots \tau_n, \quad \sigma_2 = \tau'_1 \tau'_2 \dots \tau'_m$$

Then $\text{sgn}(\sigma_1) = (-1)^n$, $\text{sgn}(\sigma_2) = (-1)^m$

$$\therefore \sigma_1 \circ \sigma_2 = \tau_1 \tau_2 \dots \tau_n \tau'_1 \tau'_2 \dots \tau'_m$$

$$\text{So } \text{sgn}(\sigma_1 \circ \sigma_2) = (-1)^{n+m} = (-1)^n (-1)^m = \text{sgn}(\sigma_1) \text{sgn}(\sigma_2)$$

Q2: Let A be a 2×2 real matrix be such that $A^2 = 0$
Show that $\det(cI - A) = c^2$ for any real number c .

Sol: Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$$\text{given } A^2 = 0 \Rightarrow \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow a_{11}^2 + a_{12}a_{21} = 0 \quad \text{--- (i)}$$

$$a_{11}a_{12} + a_{12}a_{22} = 0 \quad \text{--- (ii)}$$

$$a_{21}a_{11} + a_{22}a_{21} = 0 \quad \text{--- (iii)}$$

$$a_{21}a_{12} + a_{22}a_{22} = 0 \quad \text{--- (iv)}$$

$$\swarrow$$
$$a_{11} + a_{22} = 0$$

$$\text{Now } \det(cI - A) = \det \begin{bmatrix} c - a_{11} & -a_{12} \\ -a_{21} & c - a_{22} \end{bmatrix}$$

$$= (c - a_{11})(c - a_{22}) - a_{12}a_{21}$$

$$= c^2 - ca_{11} - ca_{22} + a_{11}a_{22} - a_{12}a_{21}$$

$$= c^2 - c(a_{11} + a_{22}) + a_{11}a_{22} - a_{12}a_{21}$$

$$= c^2 - c(a_{11} + a_{22}) + \det A$$

$$(\because \det \tilde{A} = 0)$$

$$\det A \det A = 0$$

$$\det A = 0$$

$$= c^2 - c(a_{11} + a_{22}) + 0$$

$$= c^2$$

$$(\because a_{11} + a_{22} = 0 \text{ by (ii)})$$

$$\text{hence } \det(cI - A) = c^2$$

